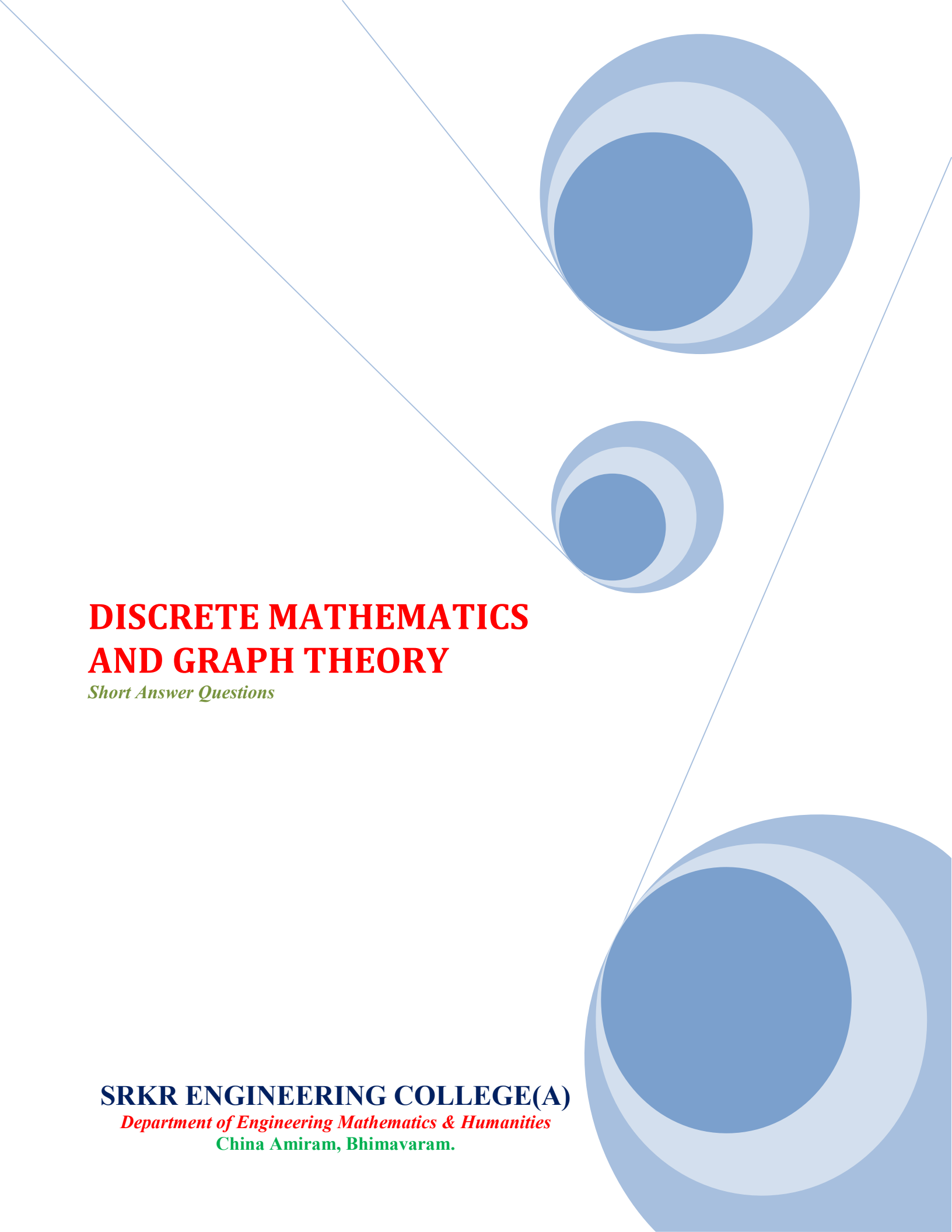




DISCRETE MATHEMATICS AND GRAPH THEORY

Short Answer Questions

SRKR ENGINEERING COLLEGE(A)

Department of Engineering Mathematics & Humanities

China Amiram, Bhimavaram.

1. MATHEMATICAL LOGIC

1. Write any three connectives with truth tables.

Ans.: **Conjunction:** Let P and Q be any two statement variables then the conjunction of P and Q is denoted by $P \wedge Q$.

Truth table:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction: Let P and Q be any two statement variables then the disjunction of P and Q is denoted by $P \vee Q$.

Truth table:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Implication: Let P and Q be any two statement variables then the implication of P and Q is denoted by $P \rightarrow Q$.

Truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

2. Construct the truth table for the statement $(P \rightarrow Q) \wedge (\neg P \rightarrow Q)$

P	Q	$\neg P$	$P \rightarrow Q$	$\neg P \rightarrow Q$	$(P \rightarrow Q) \wedge (\neg P \rightarrow Q)$
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

3. Show that $\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$ is a tautology.

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$ (A)	$(\neg P \vee \neg Q)$ (B)	$A \leftrightarrow B$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	T	T

4. Show that $(P \vee Q) \wedge (\neg P \wedge \neg Q)$ is a contradiction.

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$(\neg P \wedge \neg Q)$	$(P \vee Q) \wedge (\neg P \wedge \neg Q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

5. Prove that $\neg(P \leftrightarrow Q) \leftrightarrow (P \wedge \neg Q) \vee (\neg P \wedge Q)$

P	Q	$\neg P$	$\neg Q$	$P \leftrightarrow Q$	$\neg(P \leftrightarrow Q)$ (A)	$P \wedge \neg Q$	$\neg P \wedge Q$	$(P \wedge \neg Q) \vee (\neg P \wedge Q)$ (B)	$A \leftrightarrow B$
T	T	F	F	T	F	F	F	F	T
T	F	F	T	F	T	T	F	T	T
F	T	T	F	F	T	F	T	T	T
F	F	T	T	T	F	F	F	F	T

6. Show that $P \rightarrow (Q \rightarrow P) \Leftrightarrow \neg P \rightarrow (P \rightarrow Q)$ without using truth tables.

$$\begin{aligned}
 P \rightarrow (Q \rightarrow P) &\Leftrightarrow P \rightarrow (\neg Q \vee P) \\
 &\Leftrightarrow \neg P \vee (\neg Q \vee P) & \neg P \rightarrow (P \rightarrow Q) &\Leftrightarrow \neg P \rightarrow (\neg P \vee Q) \\
 &\Leftrightarrow (\neg P \vee \neg Q) \vee P & &\Leftrightarrow P \vee (\neg P \vee Q) \\
 \text{Sol.:} \quad &\Leftrightarrow P \vee (\neg P \vee \neg Q) & \text{and} &\Leftrightarrow (P \vee \neg P) \vee Q \\
 &\Leftrightarrow (P \vee \neg P) \vee \neg Q & &\Leftrightarrow T \vee Q \\
 &\Leftrightarrow T \vee \neg Q & &\Leftrightarrow T \\
 &\Leftrightarrow T
 \end{aligned}$$

$$\text{Hence } P \rightarrow (Q \rightarrow P) \Leftrightarrow \neg P \rightarrow (P \rightarrow Q)$$

7. Show that $P \wedge Q \Rightarrow P \rightarrow Q$

P	Q	$P \wedge Q$	$P \rightarrow Q$	$(P \wedge Q) \rightarrow (P \rightarrow Q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	F	F	T

8. Define PDNF and PCNF.

Ans.: **Principal Disjunctive Normal form (PDNF):**

An equivalence formula consisting of disjunctions of minterms only is known as its principal disjunctive normal form or Sum of products canonical form.

Principal Conjunctive Normal form (PCNF):

An equivalence formula consisting of conjunctions of maxterms only is known as its principal conjunctive normal form or Product of sums canonical form

9. Write the following in symbolic form

- i) Some men are giants. ii) All teachers are male
 iii) Some student like cricket. iv) Some monkeys have no tails.

Ans.: i) **Some men are giants.**

$M(x)$: x is a men

$G(x)$: x is gaint

The symbolic form is

$$\exists x [M(x) \wedge G(x)]$$

iii) **Some student like cricket**

ii) **All teachers are male**

$T(x)$: x is a teacher

$M(x)$: x is male

The symbolic form is

$$\forall x [T(x) \rightarrow M(x)]$$

iv) **Some monkeys have no tails.**

S(x): x is a student

C(x): x like cricket

The symbolic form is

$$\exists x [S(x) \wedge C(x)]$$

M(x): x is a monkey

T(x): x has a tail.

The symbolic form is

$$\exists x [M(x) \wedge \neg T(x)]$$

10. Write the following in symbolic form.

i) Some people who trust others one rewarded. ii) If any one good then John is good.

iii) He is ambitious or no one is ambitious. iv) It is not true that all roads lead to Rome.

Ans.: i) Some people who trust others one rewarded. ii) If any one good then John is good.

P(x): x is a person

T(x): x trust others

R(x): x is rewarded

The symbolic form is

$$\exists x [P(x) \wedge T(x) \wedge R(x)]$$

P(x): x is a person

G(x): x is good

J : John

The symbolic form is

$$\exists x [P(x) \wedge G(x)] \rightarrow J$$

iii) **He is ambitious or no one is ambitious.** iv) **It is not true that all roads lead to Rome.**

P(x): x is a person

A(x): x is ambitious

R(x): x is rewarded

The symbolic form is

$$A(y) \vee \exists x [P(x) \rightarrow \neg A(x)]$$

S(x): x is a road

L(x): x leads to roam

The symbolic form is

$$\exists x [S(x) \wedge \neg L(x)]$$

2. SET THEORY, RELATIONS, FUNCTIONS

1. *If A, B are any two sets, then $A - (A - B) = A \cap B$*

$$\begin{aligned} & A - (A - B) = A - (A \cap \overline{B}) \\ & = A \cap \overline{(A \cap \overline{B})} \\ \text{Ans.: } & = A \cap (\overline{A} \cup B) \\ & = (A \cap \overline{A}) \cup (A \cap B) \\ & = \phi \cup (A \cap B) \\ & = A \cap B \end{aligned}$$

2. *Define equivalence relation and give an example.*

Ans.: A relation R on a set X is said to be an equivalence relation if R is reflexive, symmetric and transitive.

Ex: "Is parallel to" is an equivalence relation on a set of straight lines

3. *Define Partial ordering relation and give an example.*

Ans.: A relation R on a set X is said to be a partial ordering relation if R is reflexive, anti-symmetric and transitive.

Ex: "Set Inclusion" is a partial ordering relation on a collection of sets.

4. *If R and S are transitive relations on a set A , so is $R \cap S$*

$$\begin{aligned} & \text{Let } a, b, c \in A \text{ such that } (a, b), (b, c) \in R \cap S \\ & \Rightarrow (a, b) \in R \text{ and } (b, c) \in R \text{ \& } (a, b) \in S \text{ and } (b, c) \in S \\ \text{Ans.: } & \Rightarrow (a, c) \in R \text{ and } (a, c) \in S \\ & \Rightarrow (a, c) \in R \cap S \\ & \text{Hence } R \cap S \text{ is transitive} \end{aligned}$$

5. *Suppose $A = \{1, 2, 3\}$ and $B = \{1, 2\}$ and let $R = \{(2, 1), (3, 1), (3, 2)\}$, then the relation matrix of R ?*

$$\text{Ans.: } M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

6. *Show that the relation $R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a symmetric relation on $\mathbb{R}$.*

Ans.: If $(x, y) \in R$ then $\Rightarrow x^2 + y^2 = 1$

$$\Rightarrow y^2 + x^2 = 1$$

$$\Rightarrow (y, x) \in R$$

Then R is Symmetric relation.

7. If $R=\{(1,2),(3,4),(2,2)\}$ and $S=\{(4,2),(2,5),(3,1),(1,3)\}$ the show that $RoS \neq SoR$.

Ans.: $RoS = \{(1, 5), (3, 2), (2, 5)\}$

$$SoR = \{(4, 2), (3, 2), (1, 4)\}$$

There fore $RoS \neq SoR$.

8. Let $X = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 4)\}$ be a relation on the set X . then find the Transitive closure of R

Ans.: Give $R = \{(1, 2), (2, 3), (3, 4)\}$

$$\text{Now } R^2 = RoR = \{(1, 3), (2, 4)\}$$

$$R^3 = RoRoR = \{(1, 4)\}$$

$$R^4 = \phi$$

There for the Transitive Closure of R is

$$R^+ = R \cup R^2 \cup R^3$$

$$R^+ = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}.$$

9. Find the inverse function of $f(x) = x^3 + 1$

Ans.:

$$\text{Let } y = x^3 + 1$$

$$\Rightarrow x^3 = y - 1$$

$$\Rightarrow x = (y - 1)^{1/3}$$

$$\text{Now } y = f(x) \Rightarrow x = f^{-1}(y) = (y - 1)^{1/3}$$

$$\text{Therefore, } f^{-1}(x) = (x - 1)^{1/3}$$

10. Define Bijective function and give an example.

Ans.: A function f from A to B is said to be bijective if f is both injective and surjective (i.e. both one –one and onto)

Example $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = -x$ is an example of bijective function.

11. Let f and g be two function from the positive integers to positive integers define by

$$f(n) = n^2 \text{ and } g(n) = 2^n \text{ then find } fog \text{ and } gof$$

$$(fog)n = f(g(n)) = f(2^n) = (2^n)^2 = 2^{2n}$$

Ans.: $(gof)n = g(f(n)) = g(n^2) = 2^{n^2}$

12. Define Lattice and give an example.

Ans.: A Lattice is a partially ordered set $(L, \leq)$ in which every pair of elements $a, b \in L$ has a greatest lower bound and a least upper bound.

Ex: A Poset $(Z^+, /)$ is a Lattice

13. Define distributive lattice

Ans.: A lattice $(L, \vee, \wedge)$ is called distributive lattice if for any $a, b, c \in L$ the following distributive laws holds

$$i) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$ii) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

3. COMBINATORICS

1. *In a class, there are 10 boys and 12 girls. How many ways are there to select one student to answer a question?*

Ans.: There are 10 ways to choose a boy.

There are 12 ways to choose a girl.

Since choosing a boy and choosing a girl are mutually exclusive (you can't pick both at the same time), you use the Addition Principle: $10 + 12 = 22$.

Therefore there are 22 different ways to select one student to answer the question.

2. *Suppose you have 3 types of shirts (red, blue, and green) and 2 types of pants (jeans and khakis). How many different outfits can you make?*

Ans.: There are 3 choices for shirts.

There are 2 choices for pants.

Using the Counting Principle (Product rule) the total number of outfits is $3 \times 2 = 6$

Therefore there are 6 different outfits

3. *How many distinct ways can the letters in "BANANA" be arranged?*

Ans.: The word "BANANA" has 6 letters, with "A" occurring 3 times and "N" occurring 2 times.

$$\text{Total permutations} = \frac{6!}{3! \times 2!} = \frac{720}{12} = 60$$

Therefore there are 60 distinct arrangements of the word "BANANA."

4. *How many distinct necklaces can be made using 6 different beads?*

Ans.: For 6 beads, the number of distinct arrangements (considering rotation but not reflection) is $(6-1)! = 5! = 120$

Therefore there are 120 distinct necklaces that can be made with 6 different beads.

5. *How many ways can the letters in "RACER" be arranged if the letters "R" must always be together?*

Ans.: Treat "RR" as a single unit.

Now we have 4 units to arrange: They are "RR", "A", "C", and "E".

The number of ways to arrange these 4 units is $4! = 24$

Therefore there are 24 ways to arrange the letters in "RACER" with the Rs together

6. How many ways can you choose 3 students from a group of 5 students?

Ans.: Here, $n = 5$ and $r = 3$

$$C(5,3) = \frac{5!}{3!(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1} = \frac{120}{6 \times 2} = 10$$

Therefore there are 10 ways to choose 3 students from a group of 5.

7. A committee of 4 members is to be formed from a group of 7 people. How many different committees can be formed?

Ans.: Here, $n = 7$ and $r = 4$

$$C(7,4) = \frac{7!}{4!(7-4)!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 3 \times 2 \times 1} = \frac{5040}{144} = 35$$

Therefore there are 35 different ways to form a committee of 4 members from a group of 7 people.

8. How many ways can you choose 2 elements from a set of 5 elements?

Ans.: Using the formula: ${}_5C_2 = \frac{5!}{2!(5-2)!} = \frac{5 \times 4}{2 \times 1} = 10$

Therefore there are 10 ways to choose 2 elements from a set of 5.

9) How many ways can you arrange the letters in "MISSISSIPPI"?

Ans.: The word "MISSISSIPPI" has 11 letters: 4 S's, 4 I's, 2 P's, and 1 M.

$$\text{The number of distinct arrangements is: } \frac{11!}{4!4!2!1!} = \frac{39916800}{24 \times 24 \times 2 \times 1} = 34650$$

Therefore there are 34,650 distinct arrangements of the letters in "MISSISSIPPI."

10. What is the coefficient of x^2yz in the expansion of $(x + y + z)^4$?

Ans.: The term x^2yz corresponds to $n_1 = 2$, $n_2 = 1$, and $n_3 = 1$ with $n = 4$

$$\text{Thus the coefficient is } P(4;2,1,1) = \frac{4!}{2!1!1!} = \frac{24}{2 \times 1 \times 1} = 12$$

Therefore the coefficient of x^2yz is 12.

11. Define Permutations

Ans.: Permutations refer to the different ways in which a set of objects can be arranged in order. The order of the objects is important in permutations.

12. Find the generating function for the sequence $A = \{a_r\}_{r=0}^{\infty}$ where $a_r = \begin{cases} 1 & \text{if } 0 \leq r \leq 2 \\ 3 & \text{if } 3 \leq r \leq 5 \\ 0 & \text{if } r \geq 6 \end{cases}$

Ans.: Generating function is $A(x) = 1 + x + x^2 + 3x^3 + 3x^4 + 3x^5$

13. Find the coefficient of X^{10} in $(1 + x + x^2 + \dots)^2$

$$(1 + x + x^2 + \dots)^2 = \frac{1}{(1 - X)^2} =$$

Ans.:

$$= \sum_{r=0}^{\infty} C(r+1, r) x^r$$

$$= \sum_{r=0}^{\infty} (r+1) x^r$$

Therefore the coefficient of x^{10} is 11.

14. Define Recurrence relation.

Ans.: A Recurrence relation is a formula that relates for any integer $n \geq 1$, the n th

Term of a sequence $A = \{a_r\}_{r=0}^{\infty}$ to one or more of its preceding terms $a_0, a_1, \dots, a_{n-1}$.

15. Solve $a_n = a_{n-1} + n$, where $a_0 = 2$ by substitution.

Ans.: For $n = 1$, we have $a_1 = a_0 + 1$

For $n = 2$, we have $a_2 = a_1 + 2 = a_0 + 1 + 2$

For $n = 3$, we have $a_3 = a_2 + 3 = a_0 + 1 + 2 + 3$

Similarly,

$$a_n = a_0 + 1 + 2 + 3 + \dots + n = 2 + \frac{n(n+1)}{2}$$

16. Write the characteristic polynomial of $a_n - 7a_{n-1} + 12a_{n-2} = 0$, for $n \geq 2$,

Ans.: The characteristic polynomial of $a_n - 7a_{n-1} + 12a_{n-2} = 0$, is

$$C(t) = t^2 - 7t + 12$$

17. Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$

Ans.: The characteristic polynomial is $C(t) = t^2 - 6t + 9 = (t - 3)^2$

Thus the general solution is $a_n = (C_1 + C_2 n)3^n$

18. Find the particular solutions of $a_n - 3a_{n-1} = 3^n$

Ans.: We have $a_n^p = An3^n$

Given that $a_n - 3a_{n-1} = 3^n \dots (1)$,

Substituting the value $a_n^p = An3^n$ in eq(1)

We get, $A = 1$

$\therefore a_n^p = n3^n$ is the particular solution.

19. Write the general form of a particular solution for $a_n - 7a_{n-1} + 12a_{n-2} = 2^n$.

Ans.: General form of particular solution is $a_n^p = A2^n$

20. Write the general form of a particular solution for $a_n - 7a_{n-1} + 12a_{n-2} = 3^n$.

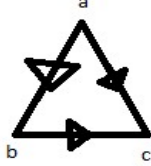
Ans.: General form of particular solution is $a_n^p = An3^n$

4. GRAPH THEORY

1. Define directed graph with example.

Ans.: If the elements of E are “ordered pairs denoted by (u, v) then G is called a directed graph or digraph.

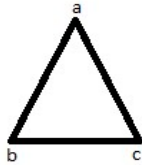
Ex: $G=(V,E)$ where $V=\{a, b, c\}$ & $E=\{(a, b),(b, c),(a, c)\}$ is a directed graph.



2. Define Non-directed graph with example.

Ans.: If the elements of E are “unordered pairs denoted by $\{u,v\}$ then G is called a non-directed graph.

Ex: $G=(V,E)$ where $V=\{a, b, c\}$ & $E=\{(a, b),(b, c),(a, c)\}$ is a non-directed graph.



3. State sum of degrees theorem.

Ans.: Let G be a graph with $|V| = n$ vertices and $|E|$ edges then $\sum_{i=1}^n \deg(v_i) = 2|E|$.

If G is a directed graph then $\sum_{i=1}^n \deg^+(v_i) = \sum \deg^-(v_i) = |E|$

4. Find the number of edges in a graph containing 2 vertices of degree 3, 3 vertices of degree 4 and a vertex of degree 6.

$$\sum_{i=1}^n \deg(v_i) = 2|E|$$

Ans.: By sum of degree theorem $2 * 3 + 3 * 4 + 1 * 6 = 2|E|$

$$6 + 12 + 6 = 2|E|$$

$$|E| = 12$$

Therefore the number of edges in G is 12.

5. Define Cycle.

Ans.: A Circuit in which all the vertices are distinct except the first and last vertices is a cycle.

6. Define connected graph with example.

Ans.: A graph is called connected, if there exists a path from a vertex to any other vertex.



7. Define Isomorphic.

Ans.: Two graphs G_1 and G_2 are isomorphic if there is a function $f: V(G_1) \rightarrow V(G_2)$ such that

- i) f is one-to-one
- ii) f is onto
- iii) f preserves adjacency

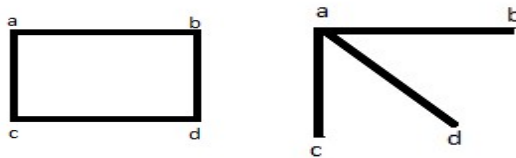
8. Define subgraph.

Ans.: A graph $H = (V_1, E_1)$ is called a Subgraph of $G = (V, E)$ if and only if $V_1 \subseteq V$ and $E_1 \subseteq E$.

9. Define Spanning Subgraph with example.

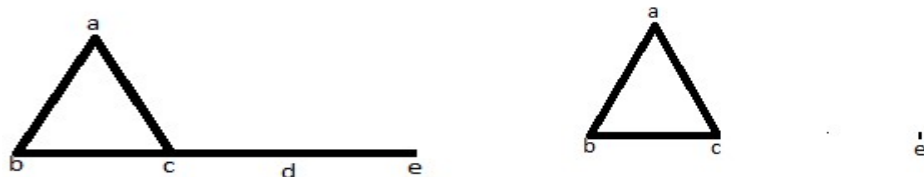
Ans.: A graph $H = (V_1, E_1)$ is called a spanning Subgraph of $G = (V, E)$ if and only if $V_1 = V$ and $E_1 \subseteq E$

Ex.



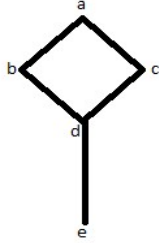
10. Define Cut-Vertex with example.

Ans.: A vertex V is called a cut-vertex or a cut-point or a connected graph G if $G - V$ is disconnected.

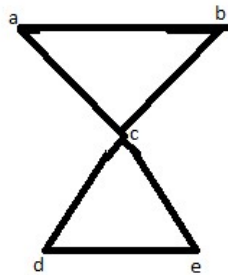


11. Define Euler path and Euler Circuit with example.

Ans.: Euler Path: A path in a graph G is called an Euler path if it includes every edge exactly once.

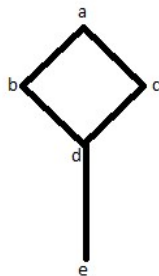


Euler Circuit: An Euler circuit is a circuit which includes every edge exactly once.

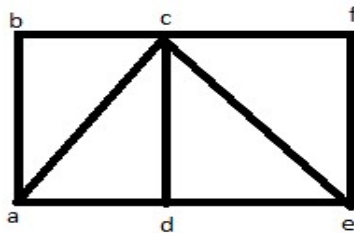


12. Define Hamilton Path & Hamilton Circuit.

Ans.: Hamilton Path: A Hamilton Path in a graph G is a path that contains each vertex exactly once.



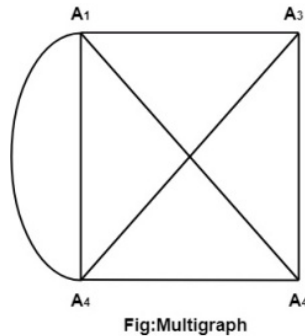
Hamilton Circuit: A Hamilton Circuit is a circuit that contains each vertex except the starting vertex exactly once.



5. MULTIGRAPHS

1. Define multigraph.

Ans.: A graph which contains some parallel edges and or self loops are called multi graph



2. Define bipartite graph.

Ans.: A graph is said to be bipartite graph if the vertex set can be partitioned into two sets M and N such that the vertices of M are adjacent to the vertices of N only

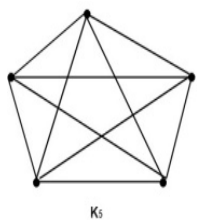
3. Define Planar Graph

Ans.: A graph is said to be planar if it can be drawn in a plane without any edge crossover.

4. Define Non-Planar Graph give an example of it.

Ans.: A graph is said to be non planar if it cannot be drawn in a plane so that no edge cross.

Example



5. State Euler's formula for planar graphs.

Ans.: In a connected planar graph G, $|V| - |E| + |R| = 2$ where $|V|$: the no of vertices

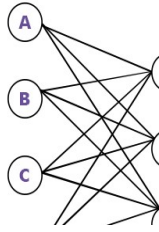
$|E|$: the no of edges , $|R|$: the no of regions.

6. Define Chromatic number of G.

Ans.: The minimum number of colors needed to produce a proper coloring of a graph G is called the chromatic number of G and is denoted by $\chi(G)$.

7. Find the Chromatic number for the following graph.

Ans.: The given graph is a bipartite graph and chromatic number $\chi(K_{m,n}) = 2$



8. Define Tree.

Ans.: A graph which has no cycle is called an acyclic graph. A tree is an acyclic graph or graph having no cycles.

A tree or general trees is defined as a non-empty finite set of elements called vertices or nodes having the property that each node can have minimum degree 1 and maximum degree n

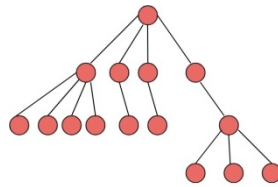


Fig:General Trees

9. Write any two properties of the trees.

Ans.: Properties of Trees:

1. There is only one path between each pair of vertices of a tree.
2. A tree with n vertices has exactly n-1 edges

10. Define minimal spanning tree.

Ans.: A minimal spanning tree is a tree such that $C(T)$ is minimal where $C(T)$ denotes the cost construction of spanning tree
